

PROJECT DEVELOPMENT PHASE

Development Environment

Platform Used

ServiceNow Developer Instance

Development Modules

- ServiceNow Studio
- Flow Designer
- Custom Tables
- Form Designer
- Business Rules
- Reports & Dashboards

Application Development

Step 1: Creation of Application

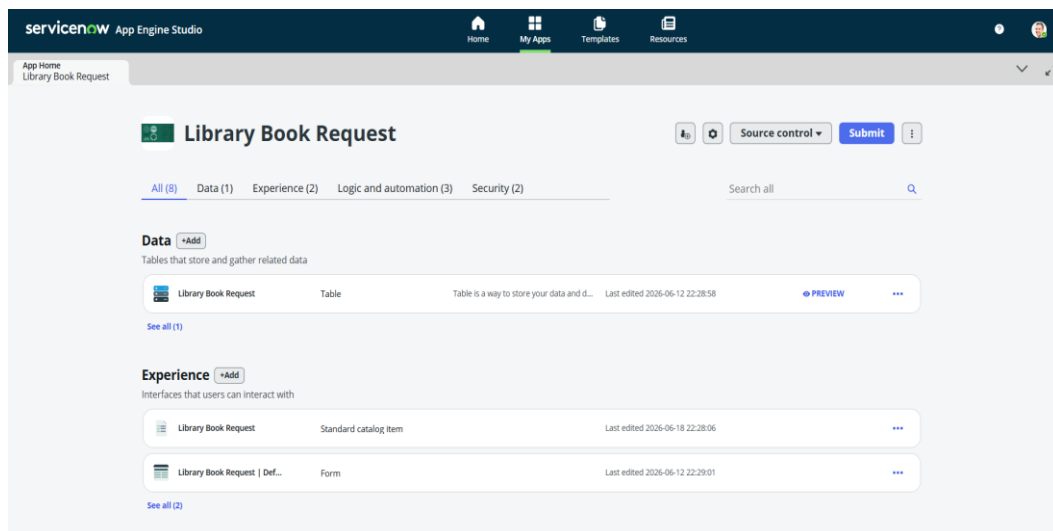
A custom application named: **Library Book Request** was created using ServiceNow Studio.

Purpose

The application serves as a centralized platform for managing student book requests from submission to librarian notification.

Application Components

- Custom Table
- Form
- Business Rules
- Flow (Notification)
- Access Controls
- Reports & Dashboard



Custom Table Development

A custom table was created to store library book request information.

Table Name

Library Book Request (u_library_book_request)

Column label * TL	Column name *	Type *	Reference	Max length	Default value	Display	Updated
Author	author	String		40		☐	2026-06-12 22:48:1
Book Title	book_title	String		40		☐	2026-06-12 22:46:5
Created	sys_created_on	Date/Time				☐	2026-06-12 22:28:5
Created by	sys_created_by	String		40		☐	2026-06-12 22:28:5
Issue Date	issue_date	Date				☐	2026-06-12 22:48:4
Librarian Notes	librarian_notes	String		40		☐	2026-06-12 22:53:3
Number	number	String		40	javascript:global.ge...	☐	2026-06-12 22:28:5
Request Number	request_number	String		40		☐	2026-06-12 22:52:1
Return Date	return_date	Date				☐	2026-06-12 22:49:2
Status	status	Choice Choices				☐	2026-06-12 22:53:3
Student Email	student_email	String		40		☐	2026-06-12 22:46:0

Fields Created

Field Name

Description

Student Name

Student requesting the book

Student Email

Student's email for contact and notification reference

Book Title

Title of the requested book

Author

Author of the requested book

Issue Date

Date the student wants the book issued

Return Date

Date the student is expected to return the book

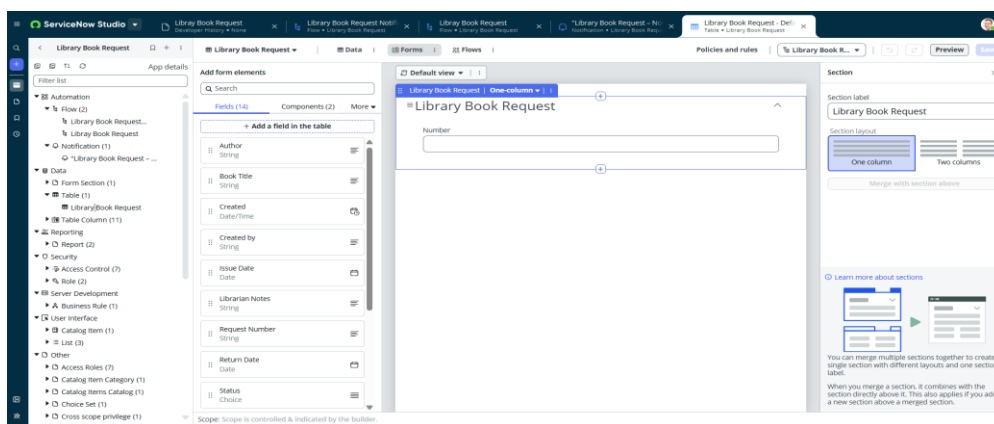
Field Name	Description
Request Number	Auto-generated unique tracking number
Status	Approval state (Pending, Approved, Rejected)
Librarian Notes	Notes added by librarian during review

Purpose

The table stores all request records and acts as the primary data repository for the application.

Form Design Development

A user-friendly form was designed for student request submission.



Form Fields

Student Information

- Student Name
- Student Email

Book Details

- Book Title
- Author

Request Tracking

- Issue Date
- Return Date
- Request Number (read-only)
- Status
- Librarian Notes

Validation

Mandatory fields and a date validation rule were configured to ensure complete and valid request submission.

Benefits

- Improved usability
- Reduced data entry errors
- Standardized request collection

Business Rule Development

Two Business Rules were developed to automate validation and tracking.

Business Rule 1 — Auto Generate Request Number

- Runs Before Insert
- Generates a unique LBR-prefixed request number for every new record

Business Rule 2 — Validate Issue Date

- Runs Before Insert and Update
- Blocks submission if the Issue Date is earlier than the current date
- Displays an error message to the student if validation fails

servicenow All Favorites History Workspaces Admin Library Book Request - Create LBR0000022

Library Book Request New record

Request Number

Student Name siva

Student Email siva@gmail.com

Book Title Atomic Habits

Author James Clear

Issue Date 2026-06-19

Return Date 2026-06-30

Status -- None --

Librarian Notes

Submit

Library Book Requests Author Search								
All								
Author	Book Title	Issue Date	Librarian Notes	Request Number	Return Date	Status	Student Email	Student Name
James Clear	Atomic Habits	2026-06-21		LBR613242	2026-06-30	Rejected	siva@gmail.com	sivapriya
Paulo Coelho	The Alchemist	2026-06-20		LBR118540	2026-06-30	Pending	test@gmail.com	Test Student
James Clear	Atomic Habits	2026-06-20		LBR462794	2026-06-30	Rejected	siva@gmail.com	siva
James Clear	Atomic Habits	2026-06-19		LBR255807	2026-06-30	Rejected	demo@gmail.com	Demo Student
Robert Kiyosaki	Rich Dad Poor Dad	2026-06-22		LBR369141	2026-07-05	Rejected	suji@gmail.com	Sujitha
APJ Abdul Kalam	Wings of Fire	2026-06-21		LBR279692	2026-07-01	Approved	priya@gmail.com	Priya
James Clear	Atomic Habits	2026-05-02		LBR701523	2026-07-20	Rejected	demo@gmail.com	demo student 2
Paulo Coelho	The Alchemist	2026-06-20		LBR087333	2026-06-30	Pending	test@gmail.com	test student 2
Paulo Coelho	The Alchemist	2026-06-19		LBR288920	2026-06-30	Pending	test@gmail.com	Test Student
James Clear	Atomic Habits	2026-06-19		LBR551226	2026-06-30	Rejected	demo@gmail.com	Demo Student 1
Paulo Coelho	The Alchemist	2026-06-20		LBR185504	2026-06-30	Pending	varshu@gamil.com	Varshini
Paulo Coelho	The Alchemist	2026-06-20		LBR087141	2026-06-30	Approved	test@gmail.com	Test Student 1

Workflow Development Using Flow Designer

A workflow was created using Flow Designer to automate the notification process.

Flow Name

Library Book Request Notification

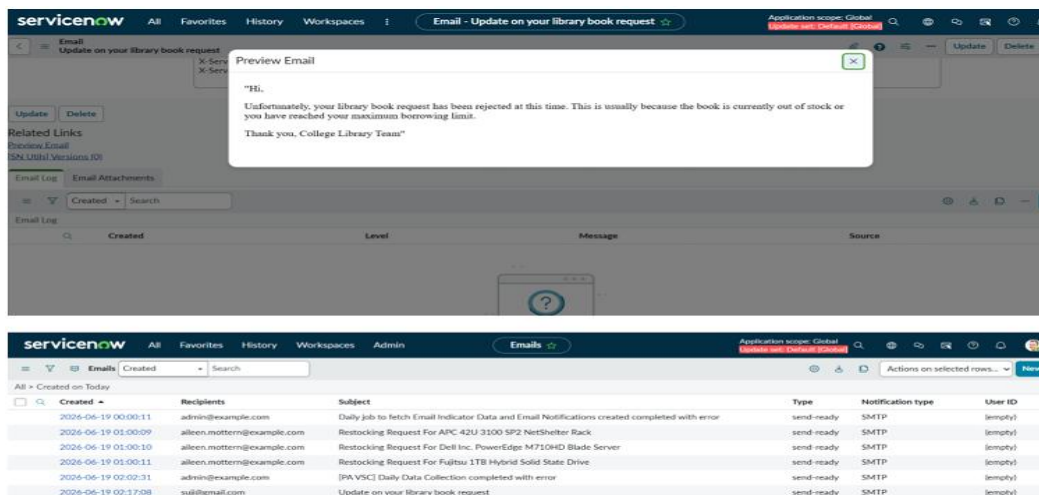
Trigger

Record Created on the Library Book Request table

The flow starts automatically whenever a new book request record is created.

Notification Action

A Send Email action was configured.



Notification Purpose

Notify the librarian that a new book request has been submitted.

Notification Content

The notification informs the librarian of:

- Request Number
- Student Name and Email
- Book Title and Author
- Issue Date and Return Date
- Current Status

EXECUTION DETAILS

Library Book Request

Test Run - Completed

Open flow

Open context record

Show Action Details

State

Start time

FLOW STATISTICS

Run as: System Administrator

Open flow logs

Completed

2026-06-19 02:17:06

1601ms

TRIGGER

Library Book Request Created

Open current record

ACTIONS

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